Mock 1 – Organic Functional Group Tests

Aim

To carry out simple test reactions to identify some important organic functional groups.

Apparatus and Chemicals

- small pieces of metallic sodium
- Fehling's solution 1
- Fehling's solution 2
- bromine water
- 2 $mol.dm^3$ ethanoic acid
- sodium hydrogencarbonate solid
- sodium hydroxide solution
- silver nitrate solution
- dilute nitric acid
- acidified potassium dichromate(VI) solution
- acidified potassium manganate(VII) solution
- anti-bumping granules
- ethanol
- propanal
- cyclohexene
- limonene
- 1-bromobutane
- 250cm^3 beaker
- test tubes and rack
- test tube holder and bung
- dropping pipettes
- thermometer
- stop clock
- Access to a hot plate water bath

Risk Assessment

	Hazard	Risk	Control Measure
Sodium hydroxide	corrosive	Damage to skin and eyes	Lab coat, goggles and gloves
Sodium hydrogen-carbonate	irritant	Possible inhalation and lung damage	Keep solid away from face
Organic samples	flammable toxic	Possible fire Risk of inhalation	No naked flames Ventilate

Complete the risk assessment.
Sodium metal is flammable and should be handled with forceps.
Assume all other reagents are corrosive and toxic.

Test for an Alcohol

All the small alcohols are just acidic enough to react with sodium metal.

1. Using forceps, carefully take a small sample of sodium metal from its protective oil. Remove the remaining oil by rolling the sodium on a piece of paper towel.
2. Drop the small piece of metallic sodium into approximately 1cm^3 of ethanol in a test tube, (You may need to tap the open forceps on the neck of the test tube.)
3. Record your observations in the 'Results' section.
4. Leave this test tube for twenty minutes before you dispose safely any excess sodium using the beaker of ethanol provided. Rinse the test tube with tap water and put in the washing up bowl.

We can use an oxidising agent to show that ethanol is not a tertiary alcohol. Both primary and secondary alcohols can be oxidised but tertiary alcohols resist oxidation.

1. Use a dropping pipette to transfer about 1cm^3 of ethanol into a clean test tube then add an equal volume of acidified K_2CrO_7 solution.
2. Add a bung and shake well.
3. Remove the bung and place the test tube in a water bath on a hot plate at 60^oC for three minutes.
4. Record your observations.
5. Rinse the test tube with tap water and put in the washing up bowl.

Test for an Aldehyde

Of the carbonyls, only aldehydes can be easily oxidised using mild oxidising agents like Fehling's or Tollen's reagents. Ketones resist oxidation.

1. In a clean test tube mix together about 1cm^3 of Fehling's solution 1 and 1cm^3 of Fehling's solution 2. Use a bung to help mix. (The resultant Fehling's test reagent should be a clear dark blue solution.)
2. Add 10 drops of this test reagent to 1cm^3 of propanal in a test tube with a few anti-bumping granules.
3. Warm the test tube at 60^oC for about two minutes in a water bath.
4. Gradually bring the temperature up on the hot plate until the water just starts to boil.
5. Gently boil for a further two minutes then reduce the temperature back to 60^oC.
6. Using the test tube holder, carefully lift the test tube out of the boiling water and allow its contents to stand for two more minutes.
7. Record your observations.
8. Rinse the test tube with tap water and put in the washing up bowl.

Test for an Alkene

Bromine water can be used as a classic test for unsaturation.

1. In the fume cupboard, put about $1cm^3$ of cyclohexene in a test tube then add an equal volume of bromine water.
2. Add a bung and shake well.
3. Record your observations.
4. Make sure that you dispose the products safely in the large waste beaker provided in the fume cupboard. Rinse the test tube with tap water and put in the washing up bowl.

We can also use a strong oxidising agent like acidified potassium manganate(VII) as an alternative test for unsaturation, where an alkene becomes a diol.

1. To approximately $1cm^3$ of limonene in a clean test tube, add an equal volume of acidified potassium manganate(VII) solution.
2. Add a bung and shake well.
3. Record your observations.
4. Make sure that you dispose the products safely in the large waste beaker provided in the fume cupboard. Rinse the test tube with tap water and put in the washing up bowl.

Test for a Carboxylic Acid

The traditional test for organic acids is a reaction with a carbonate or hydrogencarbonate.

1. Place approximately $2cm^3$ of dilute ethanoic acid in a clean test tube.
2. Add one small spatula measure of solid sodium hydrogencarbonate.
3. Record your observations.
4. Rinse the test tube with tap water and put in the washing up bowl.

Test for a Halogenoalkane

If we hydrolyse a halogenoalkane then the released halide ion can be tested with silver nitrate solution. The precipitate is then used to identify the original halogen.

1. Using a dropping pipette, add approximately 10 drops of 1-bromobutane to about $1cm^3$ of sodium hydroxide solution in a test tube.
2. Add a bung and shake well.
3. Remove the bung and warm the contents of the test tube at 60^oC for five minutes in a water bath.
4. Use the test tube holder to remove the test tube then acidify the contents by adding $2cm^3$ of dilute nitric acid.
5. Add $1cm^3$ of silver nitrate solution.
6. Record your observations.
7. Rinse the test tube with tap water and put in the washing up bowl.

Results

1. Test for an Alcohol

Observations with sodium metal

Sodium floats on ethanol and fizzes. Evolution of bubbles of a colourless gas.
The metal slowly 'dissolves' away.
Alkali metals are all reactive and low density. Ethanol is just acidic enough to have its proton reduced by sodium's electron releasing hydrogen gas. A colourless solution of sodium ethoxide is also made.

Observations with acidified potassium dichromate(VI)

Orange dichromate(VI) solution becomes green.
Cr(VI) is reduced down to Cr(III) while ethanol is oxidised to ethanal and then ethanoic acid.

2. Test for an Aldehyde

Observations with Fehling's reagent

Deep blue solution turns into a cloudy reddish-brown precipitate.
Cu(II) is reduced to Cu(I) while the propanal is oxidised to propanoic acid.

3. Test for an Alkene

Observations with bromine water

Orange bromine water is decolourised. Two immiscible liquid layers form.
Various bromoalkanes result which are insoluble in the remaining aqueous layer.

Observations with acidified potassium manganate(VII)

Purple manganate(VII) solution first turns into a cloudy brown suspension then slowly forms a decolourised solution.
Various diols result which are just about soluble in the remaining aqueous layer.
Mn(VII) is reduced down to first Mn(IV) then Mn(II).

4.. Test for a Carboxylic Acid

Observations with sodium hydrogencarbonate

Lots of effervescence as bubbles of a colourless gas are evolved.
Ethanoic acid is a good enough acid to displace carbonic acid from its salt i.e. CO_2 and H_2O are made which 'adds up' to carbonic acid, H_2CO_3.

5. Test for a Halogenoalkane

Observations with acidified silver nitrate

A faint pale cream precipitate forms.
The halogenoalkane is first hydrolysed by the NaOH into an alcohol and the halide ion liberated into solution. The HNO_3 removes excess OH^- ions (which would otherwise precipitate out with the Ag^+ ions.)
A cream precipitate of silver(I) bromide forms.

Mock 2 – Inorganic Analysis

Aim

Part One - To identify six inorganic salts by the interactions between their solutions
Part Two - Carry out some simple inorganic tests to identify some unknown solids

Apparatus and Chemicals

Part One

- test tubes
- test tube rack
- dropping pipettes

- solutions of the following salts, ***randomly labelled*** A-F :
  $Ba(NO_3)_2$
  $Pb(NO_3)_2$
  $MgSO_4$
  KI
  Na_2CO_3
  $Zn(NO_3)_2$

Part Two

- 3 boiling tubes
- boiling tube rack
- 3 spatulas
- 25cm^3 measuring cylinder
- 3 damp splints
- dropping pipettes

- Fe^{2+} or Fe^{3+}, Ca^{2+}, Na^{+} as solid salts
- SO_4^{2-}, NO_3^{-}, CO_3^{2-} as solid salts
- nitric acid solution
- barium nitrate solution
- sodium hydroxide solution

Risk Assessment

	Hazard	Risk	Control Measure
Barium nitrate	toxic	Ingesting or inhaling	Lab coat and goggles
Lead nitrate	toxic	Ingesting or inhaling	Clear spillages
Magnesium sulfate	irritant	Low risk	"
Potassium iodide	irritant	Low risk	"
Sodium carbonate	irritant	Low risk	"
Zinc nitrate	irritant	Low risk	"

Part One

1. Use solubility rules to complete the table below, to show the expected observations when each solution interacts with the other five.

	$Ba(NO_3)_2$	$Pb(NO_3)_2$	$MgSO_4$	KI	Na_2CO_3	$Zn(NO_3)_2$
$Ba(NO_3)_2$		No change	Heavy white ppt	No change	White ppt	No change
$Pb(NO_3)_2$	No change		White ppt	Yellow ppt	White ppt	No change
$MgSO_4$	Heavy white ppt	White ppt		No change	White ppt	No change
KI	No change	Yellow ppt	No change		No change	No change
Na_2CO_3	White ppt	White ppt	White ppt	No change		White ppt
$Zn(NO_3)_2$	No change	No change	No change	No change	White ppt	

2. Test approximately 2 cm^3 of each unknown solution with a few drops of each of the other solutions in turn and record your observations in this second table.

	A	B	C	D	E	F
A		No change	No change	No change	Yellow ppt	No change
B	No change		White ppt	White ppt	No change	No change
C	No change	White ppt		White ppt	White ppt	White ppt
D	No change	White ppt	White ppt		White ppt	No change
E	Yellow ppt	No change	White ppt	White ppt		No change
F	No change	No change	White ppt	No change	No change	

3. Now use both tables to identify each unknown solution.

Solution	Identity
A	KI
B	$Ba(NO_3)_2$
C	Na_2CO_3
D	$MgSO_4$
E	$Pb(NO_3)_2$
F	$Zn(NO_3)_2$

Part Two

You are provided with three unknown solid inorganic salts, X, Y and Z. Each salt contains one cation and one anion only. The three cations are one of Fe^{2+}, Fe^{3+}, Ca^{2+} and Na^{+}. The three anions are one of SO_4^{2-}, NO_3^{-} and CO_3^{2-}.
You need to identify them correctly.

Dissolve a level spatula of solid X in 15cm^3 of nitric acid, HNO_3, using a boiling tube. Observe what happens then use your new solution to carry out the following simple inorganic tests. Repeat with solid Y and finally solid Z. Identify each solid.

1. Flame Tests for Metal Cations
Dip a damp splint into the unknown solution, leave it a few seconds then burn in a blue bunsen flame.

Cation	Colour
Sodium	**Yellow-orange**
Potassium	**Lilac**
Calcium	**Brick red**
Strontium	**Crimson**
Barium	**Apple green**
Copper(II)	**Blue-green**

2. Test for a Sulfate, SO_4^{2-}
Add a little $Ba(NO_3)_2$ solution to 1cm^3 of the unknown. A white precipitate of $BaSO_4$ forming will indicate that a sulfate is present.

3. Test for a Carbonate, CO_3^{2-}
Add some HNO_3 solution to the unknown **solid**. Bubbles of a colourless gas indicate a carbonate is present.

4. Test for Iron(II), Fe^{2+}
Add about 1cm^3 of dilute sodium hydroxide solution to an equal volume of the unknown solution. A dark green gelatinous precipitate forms if Fe^{2+} is present (which turns dark orange on standing.)

5. Test for Iron(III), Fe^{3+}
Add about 1cm^3 of dilute sodium hydroxide solution to an equal volume of the unknown solution. A dark orange-brown gelatinous precipitate forms if Fe^{3+} is present.

Results

Test	Observations
Flame Tests for Metal Cations	**Y gives a yellow-orange flame test.** **Z gives a brick-red flame test.**
Test for a Sulfate, SO_4^{2-}	**X gives a white precipitate.**
Test for a Carbonate, CO_3^{2-}	**Y gives bubbles of a colourless gas.**
Test for Iron(II), Fe^{2+}	**X gives a dark green gelatinous precipitate.**
Test for Iron(III), Fe^{3+}	**None give a dark orange-brown gelatinous precipitate.**

Conclusions

Solid X is $FeSO_4$

Solid Y is Na_2CO_3

Solid Z is $Ca(NO_3)_2$

Solubility Rules

These rules are essential and must be memorised.

The solubility of a compound in water gives us a major clue when trying to identify an anion or cation. They are used to solve many types of analytical problems and are essential for identifying precipitates and writing ionic equations.

Salt	Soluble	Insoluble
carbonates	$(NH_4)_2CO_3$ all group I	all the others
chlorides bromides iodides	all the rest	Pb^{2+} Ag^+ Cu^+
hydroxides	NH_4OH all group I $Ba(OH)_2$ $Sr(OH)_2$	all the others
sulfates	all the rest	$BaSO_4$ $PbSO_4$ $SrSO_4$ Ag_2SO_4 $CaSO_4$ (slightly soluble)
nitrates	all	none
ethanoates	all	none

Solubility is linked to concentration. For example, calcium hydroxide can be described as sparingly soluble and so depending on the concentration you may see a faint white precipitate. Hence you may see it written as $Ca(OH)_2$ (aq) or $Ca(OH)_2$ (s). Take care to read the details in the question regarding the actual concentrations of the solutions used. Temperature is another obvious factor with most solids (but not all) more soluble at higher temperatures.

Mock 3 – Kinetics 1

Aim

Part One - To consider the possible disproportionation of thiosulfate ions in acid solution.
Part Two - An investigation of the rate of disproportionation of thiosulfate ions in acid solution.

Apparatus and Chemicals

Part Two

- $10cm^3$ measuring cylinder
- $50cm^3$ measuring cylinder
- $100cm^3$ conical flask
- $100cm^3$ beaker
- $250cm^3$ beaker
- stopclock
- white paper
- marker pen
- dropping pipettes
- water bottle

- 0.25 $mol.dm^{-3}$ $Na_2S_2O_3$ solution
- 2.0 $mol.dm^{-3}$ HCl solution

$Na_2S_2O_3$ solution

HCl solution

Risk Assessment

Hazard	Risk	Control Measure
toxic	Inhalation	Good ventilation.
corrosive	Splashes to skin and eyes	Lab. coat, goggles and gloves.

Part One - Predicting

1. Use an example reaction to explain the meaning of 'disproportionation.'

A reaction where the same element (in a species) is simultaneously oxidised and reduced.

e.g. $\overbrace{\underbrace{\overset{0}{Cl_2(g)} + H_2O(l) \rightleftharpoons \overset{-1}{HCl(aq)}}_{\text{Reduced}} + \overset{+1}{HClO(aq)}}^{\text{Oxidised}}$

2. Use this SEP data to answer (a) and (b) below.

①	$2H^+(aq) + 2SO_2(aq) + 4e^- \rightleftharpoons S_2O_3^{2-}(aq) + H_2O(l)$		$-1.15V$
②	$S_4O_6^{2-}(aq) + 2e^- \rightleftharpoons 2S_2O_3^{2-}(aq)$	$\times 2$ for (b)	$+0.09V$
③	$S_2O_3^{2-}(aq) + 6H^+(aq) + 4e^- \rightleftharpoons 2S(s) + 3H_2O(l)$		$+0.47V$

(a) Combine the first and third half-reactions above to give an overall equation for the disproportionation of thiosulfate ions.
Explain why this reaction is thermodynamically feasible.

Electrons balance and cancel.

$2S_2O_3^{2-} + \cancel{H_2O} + \cancel{6}H^+ \rightarrow 2SO_2 + \cancel{2H^+} + 2S + \cancel{3}H_2O$

$\cancel{2}S_2O_3^{2-} + \cancel{4}H^+ \rightarrow \cancel{2}SO_2 + \cancel{2}S + \cancel{2}H_2O$

$S_2O_3^{2-} + 2H^+ \rightarrow SO_2 + S + H_2O$

With a more negative SEP, the first half-cell is a 'good enough reducing agent' to reduce the third half-cell.
A cell voltage of $+1.62V$ is a good indicator of feasibility. This reaction is possible and likely!

(b) Combine the second and third half-reactions above to give another overall equation for the disproportionation of thiosulfate ions.
Explain why this second reaction is also thermodynamically feasible.

$5S_2O_3^{2-} + 6H^+ \rightarrow 2S_4O_6^{2-} + 2S + 3H_2O$.

With a more negative (less positive) SEP, the second half-cell is a 'good enough reducing agent' to reduce the third half-cell.
A cell voltage of $+0.38V$ is a good indicator of feasibility. This reaction is possible and likely.

3. Which of the two reactions that you have written would you consider to be the most likely to occur in practice? Carefully justify your answer.

Both are thermodynamically feasible with good cell voltages.
However a higher cell emf indicates greater feasibility and greater (more negative) ΔG.
Combine the first and third half-reactions to give the most feasible reaction.

$$S_2O_3^{2-} + 2H^+ \rightarrow SO_2 + S + H_2O.$$

4. Predict and explain what would happen to your chosen reaction in 3. above if the concentration of thiosulfate is raised above normal SEP conditions of 1.0 mol.dm³.

Le Chatelier's principle says that an increase in $[S_2O_3^{2-}]$ will shift half-reaction ① to the left i.e. make it a better reducing agent. It's $E^\ominus$ will become more negative than -1.15V
Similarly, half-reaction ③ will shift to the right i.e. making it a better oxidising agent. It's $E^\ominus$ will become more positive than +0.47V.
The difference between the two SEP's gets greater so overall $E^\ominus_{cell}$ gets greater.
More voltage means more negative ΔG.
The reaction becomes even more thermodynamically feasible and very likely to happen.

Or Alternate reasoning with oxidising half-cell.

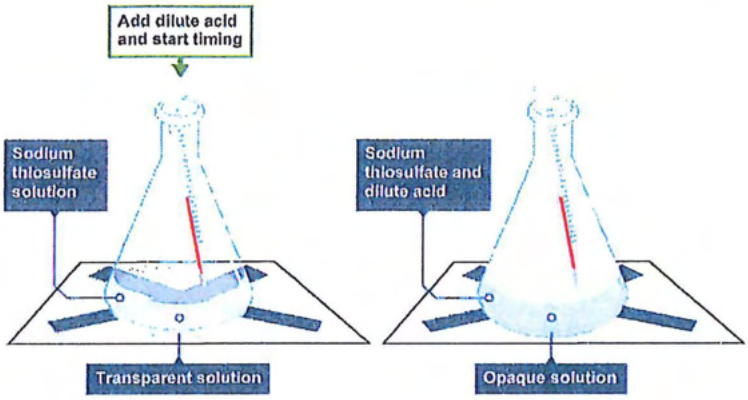
Part Two – Carrying Out

Now that you have predicted the likelihood of thiosulfate disproportionating, you need to investigate the effect of changing thiosulfate concentration on the rate of the reaction.

As we have seen, the disproportionation reaction that is most thermodynamically favourable for this reaction is

$$Na_2S_2O_{3\,(aq)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + H_2O_{(l)} + S_{(s)} + SO_{2\,(g)}$$

During the reaction the sulfur produced forms as a pale yellow suspension. This can be used to slowly obscure a cross viewed from above as shown below.



Procedure

1. Use the marker pen to draw a cross on a piece of white paper and put the conical flask on top of the cross.
2. Use the large measuring cylinder to add 50cm^3 of the sodium thiosulfate solution from the large beaker into the conical flask. A dropping pipette will help.
3. Use the small measuring cylinder to measure out 5cm^3 of the hydrochloric acid solution from the small beaker. A dropping pipette will help.
4. Now add the hydrochloric acid to the flask, swirl and start the stopclock.
5. Observe the cross from above and time how long it takes for the cross to 'disappear.' Record the time taken to the nearest second.
6. Rinse the flask with tap water then deionised water.
7. Repeat steps 2 to 6 with different concentrations of thiosulfate. Use the large measuring cylinder to make these concentrations by diluting the given solution with deionised water according to the table on the next page.
8. Record the time taken for each concentration of thiosulfate in the table.

Results

Exp No.	Volume of thiosulfate (cm^3)	Volume of water (cm^3)	Total volume (cm^3)	Initial [thiosulfate] ($mol.dm^{-3}$)	Time (s)	Rate $\times 10^{-3}$ (s^{-1})
1	50	0	50	0.250	15	66.7
2	40	10	50	0.200	21	47.6
3	30	20	50	0.150	27	37.0
4	20	30	50	0.100	42	23.8
5	10	40	50	0.050	81	12.4

Analysis

1. Calculate the missing concentrations to 3 d.p. using

$$\text{concentration of thiosulfate} = \frac{\text{volume of thiosulfate} \times 0.25}{50}$$

2. Add these concentrations to the results table then also fill in the 'Rate' column.

3. Now plot a graph of initial concentration of thiosulfate (x axis) against rate (y axis) using the graph paper on the next page.

4. Use a line of best fit and deduce the order of this disproportionation reaction with respect to thiosulfate concentration. Give your reasoning here and a rate equation.

Graph is a straight line passing through the origin so Rate $\propto [S_2O_3^{2-}]$ or Rate $= k[S_2O_3^{2-}]$
(Note that [HCl] is in large excess.)

5. Calculate the gradient of your line of best fit and record it here. Annotate your graph accordingly.

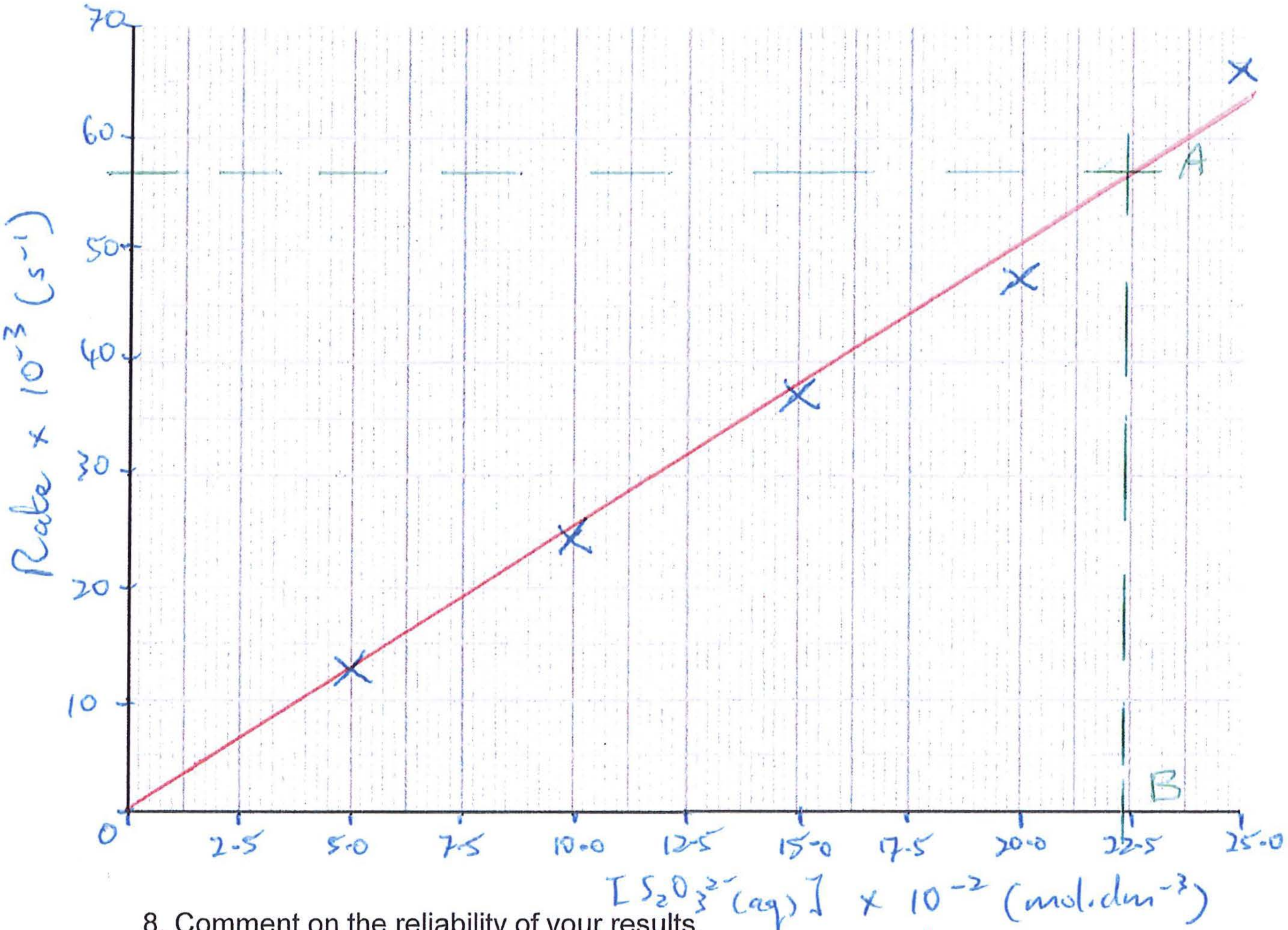
Gradient $= \frac{AB}{OB} = \frac{57 \times 10^{-3}}{22.5 \times 10^{-2}} = 0.25\ mol^{-1} dm^{+3} s^{-1}$

6. What does this gradient represent?

The rate constant, k.

7. What would happen to the gradient if the temperature is increased and why?

The gradient would increase as k increases with temperature
($k = Ae^{-RT/Ea}$ etc)



8. Comment on the reliability of your results.

Points are close to the line of best fit. Considering time was measured to the nearest second, the results are reliable.

9. How might you improve reliability?

- Repeat and use a mean.
- Use colorimetry.

10. Which piece of apparatus had the greatest uncertainty or potential error?

The stopclock (due to human reaction speed.)

11. The same reaction was found to be first order with respect to [H^+(aq)].

Give the full rate equation, deduce the units of k and then suggest a two-step mechanism for this reaction. (Assume the RDS is the first step.)

Rate = $k[S_2O_3^{2-}]^1[H^+]^1$

STEP 1 - $S_2O_3^{2-} + H^+ \rightarrow HS_2O_3^-$ RDS (slow)

STEP 2 - $HS_2O_3^- + H^+ \rightarrow SO_2 + S + H_2O$ (Fast)

Mock 4 – Kinetics 2

Aim

To investigate the effect of changing iodide concentration on the kinetics of the oxidation of iodide ions by hydrogen peroxide in acid solution

Apparatus and Chemicals

- $250cm^3$ flask
- stopclock
- 5 communal burettes, clamps and stands
- boiling tube and rack
- graduated dropping pipette

- $0.1\ mol.dm^{-3}$ H_2O_2 solution
- $1.0\ mol.dm^{-3}$ H_2SO_4 solution
- $0.1\ mol.dm^{-3}$ KI solution
- $0.005\ mol.dm^{-3}$ $Na_2S_2O_3$ solution
- deionised water
- starch solution

Risk Assessment

	Hazard	Risk	Control Measure
H_2O_2 solution	oxidising	Harms skin and eyes	Lab coat and goggles
$Na_2S_2O_3$ solution	harmful	Toxic if inhaled	Wipe up spillages
H_2SO_4 solution	corrosive	Irritant to skin	Use gloves

Background Information

Iodide ions react with hydrogen peroxide in the presence of an acid catalyst to produce iodine and water.

$$2H^+ + 2I^- + H_2O_2 \longrightarrow I_2 + 2H_2O$$

As time progresses the colour of the solution would gradually change from colourless, through yellow to orange / brown as the iodine concentration increases.

If starch was added initially however, this would not be seen. The starch would react immediately with the first iodine produced to form a dark, blue-black complex.

If we wish to measure the rate of this reaction by using this method, we have to introduce a time delay. If sodium thiosulfate is added, the thiosulfate will react with the iodine before it reacts with the starch. This is a good example of the 'iodine clock' method of following the kinetics of a reaction.

$$I_2 + 2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2I^-$$

As long as thiosulfate is present, no blue-black colour will appear.
The rate of reaction will then be given as

$$\text{rate} = \frac{\Delta [I_2]}{t}$$

If we keep the volume of the thiosulfate solution constant, the solution with change colour when the same increase in concentration of iodine is achieved. This means that

$$\Delta [I_2] \text{ is constant and therefore rate} = \frac{1}{t}$$

GC HK XAM Part of IGC HK

Procedure

1. Use the communal burettes to prepare the reaction mixture.
   Carefully add the following reagents to the 250cm^3 flask.
   - 10.0 cm^3 H_2SO_4 solution
   - 10.0 cm^3 $Na_2S_2O_3$ solution
   - 10.0 cm^3 H_2O_2 solution
   - 9.0 cm^3 deionised water
2. Use the graduated dropping pipette to add 1.0 cm^3 of starch solution to the flask.
3. Measure out 10 cm^3 of the KI solution into the boiling tube and get ready with the stopclock.
4. Pour the contents of the boiling tube into the flask, swirl once then quickly start the stopclock.
5. Time how many seconds it takes for the flask to go blue-black. Do not touch the flask.
6. Repeat steps 1 to 5 but vary the concentration of the KI by diluting with water.
   Use the boiling tube to make your four dilutions then fill in the results table for each run.

Results

Exp No.	Volume of 0.1 mol.dm^{-3} KI (cm^3)	Volume of water (cm^3)	Total volume (cm^3)	Initial [KI] (mol.dm^{-3})	Time (s)	Rate x 10^{-3} (s^{-1})
1	10	0	50	0.020	56	18
2	8	2	50	0.016	73	14
3	6	4	50	0.012	102	10
4	4	6	50	0.008	150	7
5	2	8	50	0.004	258	4

Analysis

1. Plot concentration of KI (x axis) against rate (y axis.) Draw a straight line of best fit passing through the origin..

2. Describe the relationship between KI concentration and the rate of reaction. Give a rate equation.

   Straight line passing through the origin indicates a directly proportional relationship. For example, as the concentration of KI is doubled then the rate also doubles. The reaction is first order with respect to KI concentration.

3. Find k and give its units.
   'k' is the rate constant and is found from the gradient.

$$\frac{BC}{AB} = \frac{12.2 \times 10^{-3}\ mol.dm^{-3}}{14.0 \times 10^{-3}\ s^{-1}} = \mathbf{0.87\ mol^{-1}.dm^{+3}.s^{-1}}$$

4. Suggest two other methods, not using thiosulfate, by which the rate could be measured.

   Colorimetry / light gate / conductivity measurement

5. State the largest source of uncertainty in the experiment.

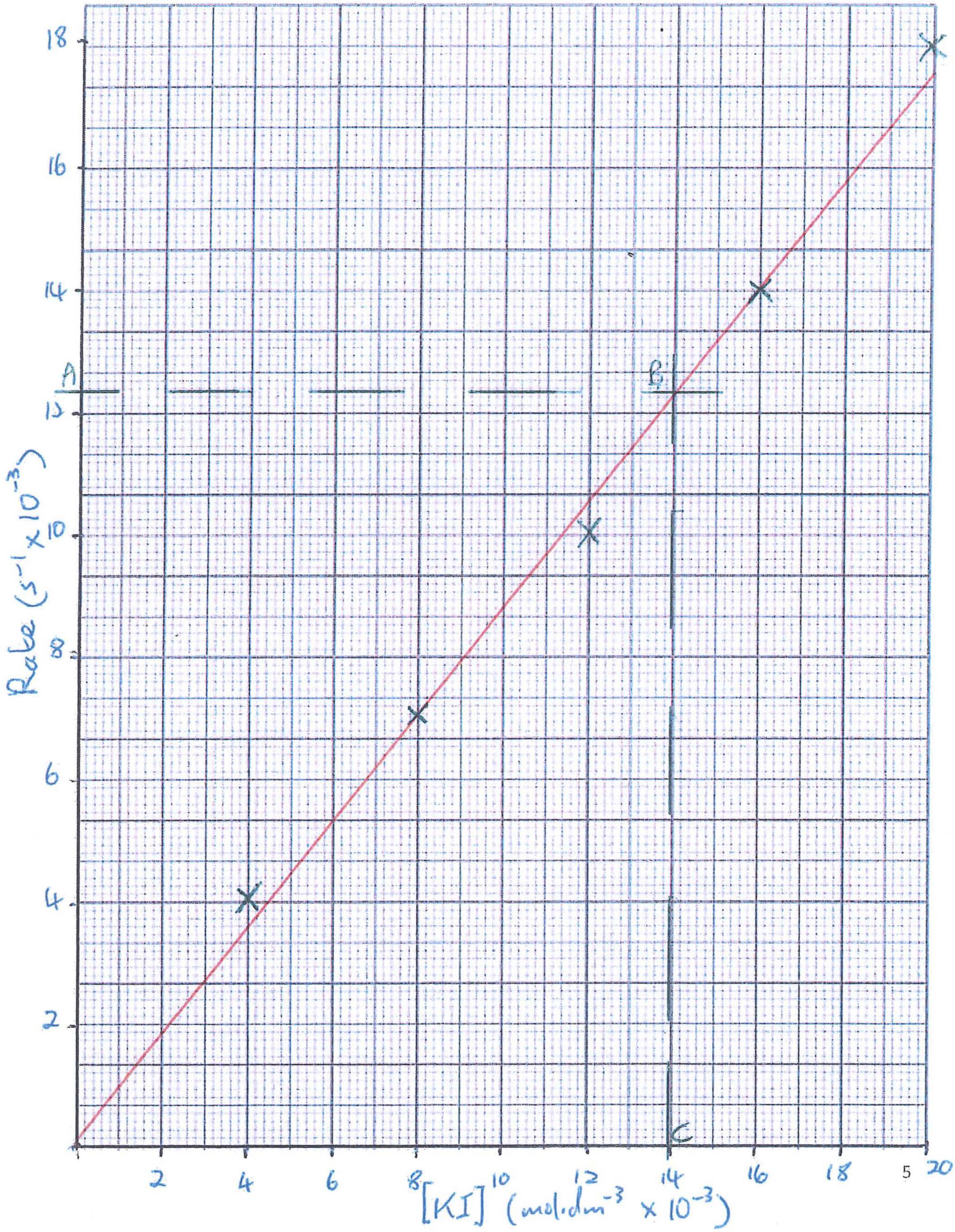
   'Uncertainties' are about potential error due to **equipment and apparatus.**
   The stopclock only measures to the nearest whole second and not to its actual resolution of 0.01 second. This is done to the limitation of human reaction time so the measurement of time is the most likely source of potential error in the experiment.

6. Suggest how the precision of the experiment could be improved.

   'Precision' is about **how fine** you can **measure a quantity** i.e. how good is the apparatus in measuring very small amounts. Difficult to improve on the burettes as they measure down to the nearest 0.05 cm^3. Easy to improve on the very approximate dropping pipette by using a much better 1 cm^3 syringe for the starch. Also, get rid of human reaction time problems by using a light gate and a computer to time the appearance of the blue-black complex. Colorimetry would similarly give a much more precise endpoint.

7. Suggest how the reliability of the experiment could be improved.

   'Reliability' is about **data** and how much **trust** we can place in it.
   Simple answer here is to carry out repeats and take a mean.
   Increasing the range of concentrations also helps in having greater faith in the results.


Rate (s^{-1} x 10^{-3})
[KI] (mol.dm^{-3} x 10^{-3})
A
B
C
2
4
6
8
10
12
14
16
18
20

Mock 5 – Kinetics 3 (Theoretical)

Aim

To investigate the reaction between hydrogen peroxide and iodide ions in acid solution.

$$H_2O_2 \; + \; 2H^+ \; + \; 2I^- \longrightarrow 2H_2O \; + \; I_2$$

Apparatus and Chemicals

- 250 cm^3 flask
- stopclock
- 50 cm^3 measuring cylinder
- 2 x 10 cm^3 measuring cylinders
- 5 cm^3 measuring cylinder
- 1 cm^3 syringe
- white tile

- 0.040 $mol.dm^{-3}$ KI solution
- 0.500 $mol.dm^{-3}$ HCl solution
- 0.005 $mol.dm^{-3}$ $Na_2S_2O_3$ solution
- 0.030 $mol.dm^{-3}$ H_2O_2 solution
- deionised water (wash bottle)
- starch solution

Risk Assessment

	Hazard	Risk	Control Measure
H_2O_2 solution	oxidising	Harms skin and eyes	Lab coat and goggles
$Na_2S_2O_3$ solution	toxic	Toxic if inhaled	Wipe up spillages
HCl solution	corrosive	Irritant to skin	Use gloves

Procedure

1. Use the appropriate measuring cylinders to prepare the reaction mixture. Carefully add the following reagents to the 250cm³ flask on the white tile.

- 40 cm^3 KI solution solution
- 10 cm^3 HCl solution
- 5 cm^3 $Na_2S_2O_3$ solution

2. Use the syringe to add 1.0 cm^3 of starch solution to the flask.

3. Use the second 10 cm^3 measuring cylinder to measure out 10 cm^3 of the H_2O_2 solution.

4. Get ready with the stopclock then add the H_2O_2 solution to the flask.

5. Swirl once then start the stopclock. Do not now touch the flask after this.

6. Time how many seconds it takes for the flask to go blue-black. Do not touch the flask.

7. Repeat steps 1 to 6 but vary the concentration of the KI, H_2O_2 and HCl by diluting in the measuring cylinders with water as outlined in the results table below.

Results Table 1

	Volume of Solution (cm^{-3})							
Exp	**KI**	**H_2O_2**	**H^+**	**Starch**	**H_2O**	**Total**	**Time (s) (model results)**	**Rate x 10^{-2} (s^{-1})**
1	40	10	10	1	0	66	60	1.67
2	30	10	10	1	10	66	80	1.25
3	20	10	10	1	20	66	120	0.83
4	40	8	10	1	2	66	74	1.35
5	40	6	10	1	4	66	99	1.01
6	40	4	10	1	6	66	141	0.71
7	40	10	8	1	2	66	60	1.67
8	40	10	6	1	4	66	61	1.64

Results Table 2

	Concentration of Solution ($mol.dm^{-3}$)				
Exp	KI	H_2O_2	H^+	Rate x 10^{-2} (s^{-1})	Relative Rate
1	0.024	0.00450	0.076	1.67	2.35
2	0.018	0.00450	0.076	1.25	1.76
3	0.012	0.00450	0.076	0.83	1.17
4	0.024	0.00364	0.076	1.35	1.90
5	0.024	0.00273	0.076	1.01	1.43
6	0.024	0.00182	0.076	0.71	1.00
7	0.024	0.00450	0.061	1.67	2.35
8	0.024	0.00450	0.046	1.64	2.31

Choose the lowest rate as '1.00' and all faster rates are then relative to it.

Analysis

1. Use the model results to calculate 'Rate' as a reciprocal or inverse of time taken and then complete Results Table 1.

2. Calculate the concentrations of the reagents in each experiment where:

$$c = c_{stock} \times \frac{V_{solution}}{V_{total}}$$

3. Now complete Results Table 2.

4. Use the results to determine the order of reaction with respect to each of the three reactants concentration. Give your reasoning.

(a) ***KI***

Looking at experiments 1 and 3, where $[H_2O_2]$ and $[H^+]$ are constant, if $[I^-]$ is doubled then rate is doubled. The reaction is first order with respect to $[I^-]$.

(b) $\boldsymbol{H_2O_2}$

Looking at experiments 4 and 6, where $[I^-]$ and $[H^+]$ are constant, if $[H_2O_2]$ is doubled then rate is doubled. The reaction is first order with respect to $[H_2O_2]$.

(c) $\boldsymbol{H^+}$

Looking at experiments 7 and 8, where $[I^-]$ and $[H_2O_2]$ are constant, if [H+] is increased by x 1.33 then rate remains unchanged. The reaction is zeroth order with respect to $[H^+]$.

5. Give the overall rate equation for this reaction and the overall order of reaction.

Rate = $k[I^-]^1[H_2O_2]^1[H^+]^0$ or Rate = $k[I^-][H_2O_2]$ Overall, 1 + 1 + 0 = 2nd

6. Use experiment 1 to calculate a value and units of the rate constant, k.

$$\text{Rate} = k[I^-]^1[H_2O_2]^1[H^+]^0$$

$$1.67 \times 10^{-3} = k \times 0.024 \times 0.0045 \times 1$$

$$k = \frac{1.67 \times 10^{-2}}{1.08 \times 10^{-4}} \qquad \frac{s^{-1}}{mol.dm^{-3} \times mol.dm^{-3}}$$

$$= \mathbf{154.6\ mol^{-2}.dm^{+6}.s^{-1}}$$

7. Suggest a mechanism for this reaction.

If both H_2O_2 and I^- show first order kinetics then one particle of each must be involved in the important rate-determining collision i.e. the slow rate determining step or 'RDS.'
The RDS is very often the first step in a reaction mechanism.

RDS	STEP 1	$H_2O_2 + I^- \longrightarrow H_2O + \cancel{OI^-}$	SLOW
	STEP 2	$\cancel{OI^-} + H^+ \longrightarrow \cancel{HIO}$	FAST
	STEP 3	$\cancel{HIO} + H^+ + I^- \longrightarrow H_2O + I_2$	FAST

Intermediates cancel

OVERALL $\quad H_2O_2 + 2H^+ + 2I^- \longrightarrow 2H_2O + I_2$

After cancelling, the one slow step and two fast steps 'add up' to the overall stoichiometric equation.

Evaluation

1. Identify the equipment used with the largest uncertainty.
The stopclock has a good resolution of 0.01s but time is only measured to the nearest second due to human reaction time. This creates the biggest uncertainty (or potential error) with equipment.

2. How could you reduce the uncertainty of the solution volumes?
Easiest way is to use more precise equipment. Burettes have a much better resolution when measuring the volume of liquids than a simple measuring cylinder.
If a burette was unavailable then uncertainty can be reduced by using larger volumes.

3. How would you improve the reliability of the results?
Reliability is about trust in your results. A simple improvement is to repeat experiments and use a mean. Larger ranges of independent variables will also help pinpoint anomalies.

4, Why is the inverse of time an appropriate measure of rate in this investigation?
The 'iodine clock' method used here means that exactly the same amount of iodine is made in each experiment. This means that the change of iodine concentration is a constant.

$$\text{Rate} = \frac{\Delta [I_2]}{\text{time}} \qquad \text{Rate} = \frac{1}{\text{time}}$$

5. Suggest an important controlled variable that should usually be considered when carrying out kinetic investigations and how you would keep it constant.
The Arrhenius equation shows that the rate constant is affected by temperature.

$$k = A.e^{-Ea/RT}$$

Temperature, therefore, must always be kept constant in kinetics experiments unless temperature itself is being investigated.
Easiest way is to use a thermostatically controlled water bath.
A simple beaker of hot water and a thermometer can sometimes be enough.